MINOR PROJECT • COMPUTER VISION

Real-Time Sign Language Recognition

Pose and hand landmarks with temporal sequence learning

CAMERA-BASED / ISOLATED SIGN CLASSIFICATION / WEBCAM DEMO

PROJECT OVERVIEW

From movement to meaning

A camera-based prototype that recognizes one sign at a time from a fixed vocabulary using short video sequences.

3,630
video clips

61
sign classes


Illustrative visualization

HOW THE SYSTEM WORKS

225 features / frame • 30 sampled frames

01 **CAMERA**
Video or webcam

→

02 **LANDMARKS**
33 pose + 42 hand points

→

03 **SEQUENCE**
Normalize 30 frames

→

04 **LSTM**
Learn movement over time

→

05 **OUTPUT**
Smoothed sign + score

BODY POSE 99 + LEFT HAND 63 + RIGHT HAND 63 = 225 VALUES PER FRAME

DATA & IMPLEMENTATION

DATASET SNAPSHOT

- 3,630 extracted sequences across 61 classes; “Still” has 30 clips and the other classes have 60 each.

MODEL & LIVE DEMO

- MediaPipe pose and both hands produce 225 normalized features per frame.
- A two-layer LSTM classifies each sequence; a CNN-LSTM option is also implemented.
- Webcam output uses a confidence threshold and smooths recent predictions.

RESEARCH CONTEXT

Pose features are compact, but may miss facial grammar and fail under occlusion (Kalinowski & Kostek, 2026).

RESULTS & SCOPE

SAVED TRAINING PLOT

~100% **near the final epochs**
in the saved split accuracy curve

Caution: the code uses this split each epoch to schedule training and choose the checkpoint. It is not an untouched final test result.

NEXT STEPS

- Evaluate once on a separate, untouched test set.
- Test with new signers and varied recording conditions; current results do not establish signer-independent performance.

SELECTED RESEARCH

Kalinowski, M. & Kostek, B. (2026). Machine learning in sign language: A comprehensive analysis and trend survey. Computer Science Review, 60, 100895. doi:10.1016/j.cosrev.2026.100895.